

Validation against published sources



MortgageMath
Precise Loan Amortization in Python

mortgagemath 0.7.0 · rendered 2026-05-06

Every fixture in the table below reproduces **every** value its source publishes, to the cent. The library follows a strict policy: if any published value diverges, the fixture stays out. The six parameter columns show the exact LoanParams settings required to match each source — the same six knobs documented in the *at-a-glance* vignette.

Validated fixtures

Source	Loan	Day	Bal	Rnd	Cmp	Freq	ARM
Arcones SOA FM §4.1 Ex 4	\$20k / 8% / 12yr	30/360	RE	HU	A	A	—
BCC Brescia IT	\$200k / 5.65% / 10yr	30/360	RE	HU	M	M	—
Broverman MIC 2.7(a)	\$12k / 12% / 3yr	30/360	RE	HU	M	M	—
Broverman MIC 2.7(b)	\$12k / 15% / 4yr	30/360	RE	HU	M	M	—
CFPB H-25(B)	\$162k / 3.875% / 30yr	30/360	RE	HU	M	M	—
CFPB IO Sample	\$211k / 4% / 30yr	30/360	RE	HU	M	M	—
eCampus §4.3 Pearlline	\$10k / 10% / 4yr	30/360	RE	HU	A	A	—
eCampus §4.3 Erika	\$32.6k / 4.83% / 9yr	30/360	RE	UP	M	M	—
eCampus §4.3 Johnetta	\$20.2k / 3.53% / 8yr	30/360	RE	HU	M	M	—
eCampus §4.4.1 T1	\$297.5k / 3.8% / 3yr, 20yr amort	30/360	RE	HU	j ₂	Q	—
eCampus §4.4.1 renew	\$265.8k / 2.5% / 17yr	30/360	RE	HU	j ₂	Q	—
FNMA §1103	\$25M / 5.5% / 10yr, 30yr amort	A/360	RE	HU	M	M	—
FHLBB 1935 Plan A	\$3k / 6% / 139mo	30/360	CP	HU	M	M	—
Geltner Ch 20	\$1M / 12% / 30yr	30/360	CP	HU	M	M	—
Goldstein §10.3	\$563 / 12% / 5mo	30/360	CP	HU	M	M	—
JHF Flat 35	\$20M / 1.5% / 30yr	30/360	RE	HU	M	M	—
Las Positas §8.05 Ex 1	\$15k / 9% / 5yr	30/360	RE	UP	M	M	—
Las Positas §8.05 Ex 3	\$18k / 2% / 5yr	30/360	RE	UP	M	M	—
LoanKeisan JP	\$30M / 1% / 30yr	30/360	RE	DN	M	M	—
MoneyVox FR	\$10k / 5% / 1yr	30/360	RE	HU	M	M	—
MS State P3920	\$100k / 7% / 30yr	30/360	RE	HU	M	M	—
Olivier Chans T1	\$350.1k / 4.9% / 3yr, 20yr amort	30/360	RE	HU	j ₂	M	—
Olivier Chans renew	\$316.6k / 5.85% / 17yr	30/360	RE	HU	j ₂	M	—
OpenStax §6.12.110	\$132.7k / 4.8% / 30yr	30/360	RE	UP	M	M	—
OpenStax AK 6.100.1	\$18.3k / 6.75% / 4yr	30/360	RE	UP	M	M	—
OpenStax AK 6.100.2	\$41.6k / 3.9% / 6yr	30/360	RE	UP	M	M	—
OpenStax AK 6.110	\$153.9k / 4.21% / 20yr	30/360	RE	UP	M	M	—
OpenStax AK 6.114	\$159.2k / 5.75% / 30yr	30/360	RE	UP	M	M	—
OpenStax AK 6.36	\$23.7k / 4.76% / 5yr	30/360	RE	UP	M	M	—
OpenStax AK 6.78.1	\$17.9k / 7.5% / 10yr	30/360	RE	UP	M	M	—
OpenStax AK 6.78.2	\$33.8k / 4.3% / 20yr	30/360	RE	UP	M	M	—
OpenStax §6.8 car	\$28.5k / 3.99% / 5yr	30/360	RE	UP	M	M	—
OpenStax §6.8 home	\$136.7k / 5.75% / 15yr	30/360	RE	UP	M	M	—
ProEducate cap	\$65k / 10% / 30yr	30/360	RE	HU	M	M	1, cap
RBC Acc Bi-Weekly	\$350k / 5% / 25yr	30/360	RE	HU	j ₂	BW	—
Reg Z H-14	\$10k / 17.41% / 30yr	30/360	RE	HU	M	M	14
Skinner §42 Ex 1	\$1k / 6% / 15yr	30/360	RE	HU	A	A	—
Skinner §42 Ex 3 piano	\$500 / 6% / 5yr	30/360	RE	HU	A	M	—
Solution Bank IT	\$200k / 7% / 30yr	30/360	RE	HU	M	M	—
Synthetic HALF_EVEN	\$100k / 4.80% / 30yr	30/360	RE	HU	M	M	—
Synthetic HALF_UP	\$100k / 4.80% / 30yr	30/360	RE	HU	M	M	—
Synthetic ROUND_UP	\$100k / 4.80% / 30yr	30/360	RE	UP	M	M	—
TI BA II Plus	\$120k / 6.125% / 30yr	30/360	RE	HU	M	M	—
Tistory KR	\$300M / 4% / 30yr	30/360	RE	HU	M	M	—
Wikipedia mortgage calc	\$200k / 6.5% / 30yr	30/360	RE	UP	M	M	—
0% Promo	\$999 / 0% / 2yr	30/360	RE	HU	M	M	—

46 fixtures total.

Column codes. Day: 30/360 US residential, A/360 Actual/360 US commercial. Bal: RE round-each-balance, CP carry-precision. Rnd: UP ROUND_UP, DN ROUND_DOWN, HU ROUND_HALF_UP, HE ROUND_HALF_EVEN. Cmp: M monthly, j₂ semi-annual (Canadian *Interest Act* §6), A annual. Freq: M SM BW W Q A (monthly to annual). ARM: count of RateChange entries; cap indicates at least one payment_cap_factor. A dash is fully fixed-rate.

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